

FINAL REPORT

MOTIVATION

Music is a huge part of my life. I spend most of my day listening to music, and I listen to all kinds of genres from all around the world. The idea for this project actually came after Bad Bunny's Super Bowl performance, which introduced his album to a much wider audience. Watching people's connection with music made me wonder: do countries actually differ in their music taste? And if so, is that difference linked to something deeper, like how wealthy a country is or how its economy is doing? This curiosity led me to ask: can economic indicators like GDP per capita, unemployment rate, or urbanization explain the differences in what kind of music people listen to across countries? I chose these indicators because they reflect very different aspects of a country's economic reality, and I wanted to see which one, if any, has the strongest connection to music preferences. That became the core question of this project.

DATA SOURCE

I used four publicly available datasets for this project. The primary dataset is the "Top 50 Spotify Songs by Each Country" from Kaggle, which contains the top 50 songs for 20 countries along with Spotify audio features such as danceability, energy, acousticness, BPM, and more. This gave me around 1,000 song entries to work with. The remaining three datasets were downloaded from the World Bank Open Data portal: GDP per capita (current US\$) to measure the economic wealth of each country; Unemployment rate (% of total labor force) to capture economic hardship; Urbanization rate (% of population living in urban areas) to reflect lifestyle differences. All datasets were collected for the year 2019 to ensure consistency. Country names were standardized across datasets before merging. After cleaning and matching, the final dataset included 17 countries, where each row represents a country with its average audio features and economic indicators.

DATA ANALYSIS

The analysis was carried out in four main parts.

Part 1: Data Collection and Merging

The Spotify dataset was grouped by country to calculate the average audio feature values for each country. The World Bank datasets were filtered to keep only the 2019 values. Country names were lowercased and stripped to ensure consistent formatting before merging. The final dataset contained 17 countries with 13 variables.

Part 2: Exploratory Data Analysis (EDA)

I created scatterplots between each economic indicator and audio features such as danceability, valence, and energy. Trend lines and Pearson correlation coefficients were added to each plot to

visualize the direction and strength of relationships. A correlation heatmap was also generated to get an overall picture of how all variables relate to each other.

Part 3: Hypothesis Testing

I tested six hypotheses using Pearson correlation tests. Each hypothesis examined whether a specific economic indicator had a statistically significant relationship with a specific audio feature. A p-value threshold of 0.05 was used to determine statistical significance.

Part 4: Machine Learning

Three ML methods were applied. Linear Regression was used to predict danceability from the three economic indicators. K-Means Clustering grouped countries into three clusters based on their audio features. Hierarchical Clustering confirmed the groupings through a dendrogram. Finally, PCA was applied to reduce dimensionality and visualize the overall patterns in the data.

FINDINGS

HYPOTHESIS TESTING RESULTS

Out of six hypotheses tested, three were statistically significant: GDP per capita and danceability ($r=-0.52$, $p=0.033$): wealthier countries tend to have less danceable popular music. Unemployment rate and danceability ($r=0.60$, $p=0.011$): the strongest finding. Countries with higher unemployment listen to more danceable music. Unemployment rate and valence ($r=0.51$, $p=0.035$): higher unemployment is also linked to more positive and happy sounding music.

Linear Regression

The linear regression model achieved an R^2 score of 0.548, meaning it explains almost 55% of the variance in danceability using GDP, unemployment, and urbanization as predictors. Unemployment rate had the strongest positive effect, while GDP had the strongest negative effect.

Clustering

Both K-Means and Hierarchical Clustering produced consistent results. Countries grouped into three clear clusters. Latin countries (Colombia, Argentina, Bolivia, Chile, Brazil, Spain) formed the most distinct cluster, characterized by high danceability and high valence. Western countries like Canada, Australia, and Belgium formed a second cluster with lower danceability. European and Asian countries formed a third more balanced cluster.

PCA

The first two principal components explained 77.5% of the total variance. PC1 clearly separated Latin countries from the rest, confirming the clustering results.

Overall

Economic factors do have some influence on music preferences, but the effect is moderate. Cultural and regional identity, particularly being a Latin country, appears to be a much stronger driver of music taste than economic conditions alone.

Limitations and Future Work

Limitations

The most significant limitation of this project is the small sample size. After merging all datasets, only 17 countries remained. This limits the statistical power of the analysis and means the results should be interpreted carefully. With such a small sample, even moderate correlations can be heavily influenced by one or two outlier countries. Another limitation is that the Spotify dataset is from 2019, which may not reflect current music trends. Music preferences can change quickly, especially with the rise of new artists and viral moments like the one that inspired this project. Major global events can also shift both economic indicators and music preferences significantly; for example, the COVID-19 pandemic in 2020 affected GDP levels, unemployment rates, and potentially the kind of music people turned to during lockdowns. Using data from a single year makes it difficult to capture these kinds of dynamic changes. The audio features used (danceability, valence, energy) are calculated by Spotify's algorithm and may not perfectly capture what makes music culturally meaningful in each country. Also, averaging the top 50 songs per country loses a lot of detail about the diversity of music within each country. Finally, correlation does not imply causation. Even where significant relationships were found, we cannot conclude that economic conditions directly cause certain music preferences. There are many confounding factors such as cultural history, language, and media influence.

Future Work

With more time and data, this project could be extended in several ways: Including more countries, especially from Africa and the Middle East, which were missing from the current dataset. Adding more socio economic indicators such as income inequality or education level. Using time series data to see how music preferences change over time in response to economic events.